

Demand Estimation 5

PhD Industrial Organization

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Plan

1. Distinguishing models of competition
2. Supply-side moments
3. Welfare from new products: theory
4. Welfare from new products: application
5. Apple-cinnamon cheerios war
6. Main takeaways of demand estimation

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Application: distinguishing between models of competition

- **Overall strategy:**
- 1. Estimate demand
- 2. Use estimates + pricing rules implied by different models of firm conduct to get price-cost margins (PCM)
 - Challenge: costs not observed
- 3. Compare predicted PCM from different models of conduct to true PCM
 - See which model of firm conduct best matches the data.
- **Main finding:**
- Nash-Bertrand pricing best matches observed PCM

Application: distinguishing between models of competition

- **Demand side:**

- Utility model is exactly what we have seen:

$$u_{ijt} = x_{jt}\beta_i + \alpha_i p_{jt} + \zeta_{jt} + \epsilon_{ijt}$$

- **Data:**

- Use top 25 cereal brands
- Scanner data $\rightarrow$ get market shares, prices, etc.
 - Aggregate to MSA-quarter level = 1124 markets.
- Advertising data
- Cereal box characteristics (nutritional information); subjective characteristics ('mushy')
- Demographics from CPS

Application: distinguishing between models of competition

- **Supply side:**

- Profits of firm f :

$$\pi_f = \sum_{j \in \mathcal{J}_f} [(p_j - mc_j)q_j(\mathbf{p}) - FC_j]$$

- Where:

- π_f : profits of firm f
- p_j : price of product j ; $\mathbf{p}$: vector of prices
- mc_j : marginal cost
- FC_j : fixed cost
- q_j : quantity of product j (depends on all prices)
- $\mathcal{J}_f$: set of products that firm f maximizes profit over

Application: distinguishing between models of competition

- **Supply side:**
- Define **conduct structure** as $J \times J$ matrix:

$$H_{jk} = \begin{cases} 1 & \text{if } \exists f \text{ where } \{j, k\} \subset \mathcal{J}_f \\ 0 & \text{otherwise} \end{cases}$$

- Elements of H either 0 or 1
- If value of element = 1: then product j and k are priced *as if jointly owned*
- Examples:
 - Single product firm pricing: identity matrix
 - Joint profit maximization: matrix of 1's

Application: distinguishing between models of competition

- **Supply side:**

- Define $\Omega_{jk} = -\partial q_k / \partial p_j \cdot H_{jk}$
 - Recall: j is index, k is column

- First order condition of firms' profit maximization problem (bold denotes vectors):

$$\mathbf{q}(\mathbf{p}) - \Omega(\mathbf{p} - \mathbf{mc}) = 0$$

- Implies pricing equation:

$$\mathbf{p} - \mathbf{mc} = \Omega^{-1} \mathbf{q}(\mathbf{p})$$

- Important: above equation implies that given conduct structure + estimates of demand substitution $\Omega \rightarrow$ can measure price-cost margins *without observing cost data*.

Application: distinguishing between models of competition: estimation

- Only estimate demand
- Use BLP algorithm
- Instruments: Hausman instruments (prices in other markets)

TABLE VI
RESULTS FROM THE FULL MODEL^a

Variable	Means (β 's)	Standard Deviations (σ 's)	Interactions with Demographic Variables:			
			Income	Income Sq	Age	Child
Price	-27.198 (5.248)	2.453 (2.978)	315.894 (110.385)	-18.200 (5.914)	—	7.634 (2.238)
Advertising	0.020 (0.005)	—	—	—	—	—
Constant	-3.592 ^b (0.138)	0.330 (0.609)	5.482 (1.504)	—	0.204 (0.341)	—
Cal from Fat	1.146 ^b (0.128)	1.624 (2.809)	—	—	—	—
Sugar	5.742 ^b (0.581)	1.661 (5.866)	-24.931 (9.167)	—	5.105 (3.418)	—
Mushy	-0.565 ^b (0.052)	0.244 (0.623)	1.265 (0.737)	—	0.809 (0.385)	—
Fiber	1.627 ^b (0.263)	0.195 (3.541)	—	—	—	-0.110 (0.0513)
All-family	0.781 ^b (0.075)	0.1330 (1.365)	—	—	—	—
Kids	1.021 ^b (0.168)	2.031 (0.448)	—	—	—	—
Adults	1.972 ^b (0.186)	0.247 (1.636)	—	—	—	—
GMM Objective (degrees of freedom)			5.05 (8)			
MD χ^2			3472.3			
% of Price Coefficients > 0			0.7			

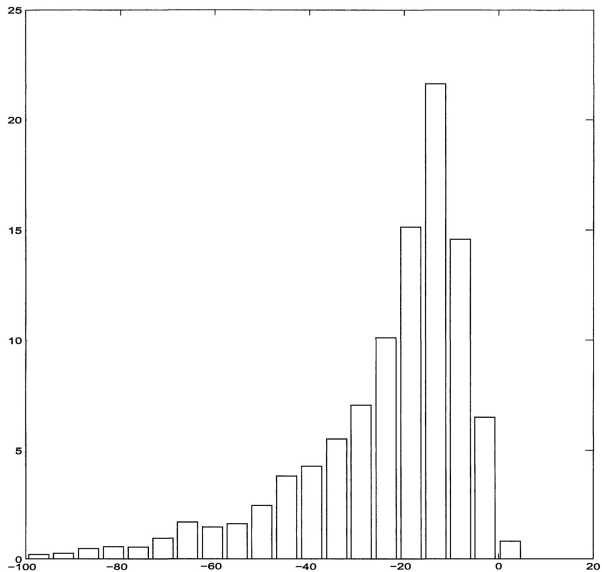


FIGURE 2.—Frequency distribution of price coefficient (based on Table VI).

TABLE VIII
MEDIAN MARGINS^a

	Logit (Table V column ix)	Full Model (Table VI)
Single Product Firms	33.6% (31.8%–35.6%)	35.8% (24.4%–46.4%)
Current Ownership of 25 Brands	35.8% (33.9%–38.0%)	42.2% (29.1%–55.8%)
Joint Ownership of 25 Brands	41.9% (39.7%–44.4%)	72.6% (62.2%–97.2%)

Application: distinguishing between models of competition: mergers

- Common use of the framework in this paper can also be used to determine the effects of a merger. How would you do this?

Application: distinguishing between models of competition: mergers

- Common use of the framework in this paper can also be used to determine the effects of a merger. How would you do this?
- Answer:
- 1. Using pre-merger data estimate demand and recover marginal costs by inverting the pricing equation:

$$\mathbf{mc} = \mathbf{p} - \Omega^{-1}\mathbf{q}(\mathbf{p})$$

- 2. Change the conduct structure H so that the merging firms jointly maximize the profits of their products

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Supply-side moments

- Previously, we used our demand model + different assumptions about how firms set prices to 'test' models of firm conduct.
- In a related idea, we could alternatively:
 - make assumptions about how firms choose prices
 - utilize data on the supply side to help identify the demand model

Supply-side moments

- Assume marginal cost is given by:

$$mc_{jt} = w_{jt}\gamma + \omega_{jt}$$

- Where:
 - w_{jt} : vector of observed characteristics of product j
 - ω_{jt} : unobserved component
 - γ : parameters to be estimated

Supply-side moments

- Also assume Nash-Bertrand pricing model and combine with cost parametrization on previous slide (using the notation from the previous lecture)

$$\mathbf{p}_t = w_t \gamma + \Omega^{-1} \mathbf{q}(\mathbf{p}_t) + \omega_t$$

- Can form supply-side moments: $E(\omega_{jt} | \mathbf{Z}_t) = 0$.
- Here: $\mathbf{Z}_t$: vector of IVs that include product characteristics, and cost shifters
- Note that above equation is informative about both supply parameters γ and the demand parameters (which affect Ω)
- Can include these additional moments in the GMM step

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Consumer welfare

- Define the **consumer surplus from a logit model**
- Expected utility prior to observing the i.i.d. logit draws from $\{1, 2, \dots, J\}$ choice alternatives: (the **'inclusive value'**)

$$\omega_{it} = E_{\{\epsilon_{i0t}, \dots, \epsilon_{iJt}\}} \max_j \{\delta_{jt} + \mu_{ijt} + \epsilon_{ijt}\} = \ln \left(\sum_j \exp\{\delta_{jt} + \mu_{ijt}\} \right)$$

- This formula is also known as the 'log-sum' formula.
- If utility is linear in price, inclusive value can be converted into dollars by dividing by the price coefficient.
- You have (probably) seen this value before, since it comes up when computing a nested logit

Consumer welfare

- Typically, two cases where we compute welfare.
- Case 1: observe quantities and prices and want to summarize them into a welfare measure.
 - Key issue: before we normalized the utility of the outside option to zero.
 - This is fine for estimating choice probabilities (why?)
 - But, issues occur if want to compute inclusive value over time (or across markets)
 - This is because we would be *implicitly assuming that utility from the outside good is constant over time*.

Consumer welfare

- Typically, two cases where we compute welfare.
- Case 1: observe quantities and prices and want to summarize them into a welfare measure.
 - Example: What if we see the share of the 'inside' products increasing over time.
 - Question: What could cause this?

Consumer welfare

- Typically, two cases where we compute welfare.
- Case 1: observe quantities and prices and want to summarize them into a welfare measure.
 - Example: What if we see the share of the 'inside' products increasing over time.
 - Question: What could cause this?
 - Answer: price of inside goods decreased (or quality ξ_{jt} increased) OR the outside option got worse. Different welfare implications, but assuming outside good is 0 rules out latter.
 - Partial solution: Nevo (2003) compute welfare over time for when ξ_{jt} changes / outside option constant vs ξ_{jt} fixed /outside option flexible. Report both extreme cases.

Consumer welfare

- Case 2: Use the model to compute a welfare gain from a counterfactual outcome.
- Assume we observe one market over time (denoted by t).
- Can show that the change in welfare from introducing a product in period t (which comes from the change in the inclusive value) is:

$$\textbf{Logit: } \ln \left(\frac{1}{s_{0t}} \right) - \ln \left(\frac{1}{s_{0t-1}} \right)$$

$$\textbf{Mixed logit: } \int \ln \left(\frac{1}{s_{i0t}} \right) dF(D_{it}, v_{it}) - \int \ln \left(\frac{1}{s_{i0t-1}} \right) dF(D_{it-1}, v_{it-1})$$

- So, logit model: welfare directly related to share of the outside good.
- Mixed logit: same idea, but difference depends on heterogeneity of choosing the outside option.

Consumer welfare

- Case 2: Use the model to compute a welfare gain from a counterfactual outcome.
- Problem with computing welfare from new goods: **red-bus blue-bus problem**.
- Thought experiment:
 - Market where consumers choose how to commute.
 - Choices: Car, Red Bus
 - Assume half consumers choose Car, half choose Red Bus
 - Assume we artificially introduce a new option: **the Blue Bus**
 - Artificial because we also **assume consumers are color-blind**
 - (Also assume price, frequency of service etc are not impacted)
 - Now, half consumers choose a car, and the rest are split between the two buses.
 - Clearly, **consumer welfare has not changed**.

Consumer welfare

- Case 2: Use the model to compute a welfare gain from a counterfactual outcome.
- **What if we now use our logit model to estimate the welfare effects of introducing a Blue Bus?**
 - (Suppose we only observe data pre-introduction of the Blue Bus.)

Consumer welfare

- Case 2: Use the model to compute a welfare gain from a counterfactual outcome.
- **What if we now use our logit model to estimate the welfare effects of introducing a Blue Bus?**
 - (Suppose we only observe data pre-introduction of the Blue Bus.)
 - Pre-introduction:
 - Assume car is outside good, normalize to zero: $\delta_{car} = 0$. Then, also, $\delta_{red-bus} = 0$ since $s_{car} = s_{red-bus} = 0$.
 - Inclusive value is: $\ln(e^0 + e^0) = \ln(2)$.
 - Post-introduction:
 - $\delta_{blue-bus} = \delta_{red-bus} = 0$ (since same bus)
 - So, $s_{car} = s_{blue-bus} = s_{red-bus} = 1/3$. Inclusive value is: $\ln(3)$, a **welfare increase!**

Consumer welfare

- Case 2: Use the model to compute a welfare gain from a counterfactual outcome.
- Where did we go wrong with the previous analysis?

Consumer welfare

- Case 2: Use the model to compute a welfare gain from a counterfactual outcome.
- Where did we go wrong with the previous analysis?
- Main issue: we are getting an extra logit draw when we introduce the Blue Bus.
- Possible solution: if we observe the market post-introduction of the new product (i.e. the Blue Bus).
- Post-introduction:
 - $s_{car} = 0.5, s_{blue-bus} = s_{red-bus} = 0.25$
 - Implies: $\delta_{blue-bus} = \delta_{red-bus} = \ln(0.5)$
 - Inclusive value: $\ln(e^0 + 2 * e^{\ln(0.5)}) = \ln(2)$. Note that this is the correct answer.

Consumer welfare

- Main takeaways from the above 'red-bus blue-bus' exercise:
- 1. Introducing new products also introduces with extra logit draws, which can bias welfare computations
- 2. Observing post-introduction market shares can 'correct' for this bias.
- Note that the above is true as well in Mixed Logit and other models:
 - Berry and Pakes (2007): "the fact that the contraction fits the shares exactly means that the extra gain from the logit errors is offset by lower δ 's, and this roughly counteracts the problems generated for welfare measurement by the model with tastes for products."

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Application of welfare from new products: the minivan (Petrin, 2002)

- **Question:** what are the welfare benefits of the minivan?
- Minivan:
- Box-like vehicle first introduced in 1984
- Increased height, decreased floor, making entry easier and allowing more movement in the vehicle
- Extremely popular with families in the 80s, 90s, etc



Application of welfare from new products: the minivan (Petrin, 2002)

- Motivation: new products, and product improvements, are a key source of economic growth. But how should we actually value these innovations?
- Main takeaways (beyond valuing the minivan):
 - 1. Logit model performs extremely poorly compared to mixed logit
 - 2. Use of micro-moments
 - 3. Dealing with the 'red-bus blue-bus' problem

Application of welfare from new products: the minivan (Petrin, 2002)

- (Conditional, indirect) utility:

$$u_{ijt} = x_{jt}\beta_{it} + \alpha_{it} \ln(y_i - p_{jt}) + \xi_{jt} + \epsilon_{ijt}$$

- Price coefficient allowed to vary by income
- Other notes: 'minivan' enters *as a characteristic*, also includes a supply side
- Data
- Observes US market 1981-1993 (for 916 vehicles)
- Prices, quantities (from Automotive Data Book)
- Product characteristics: fuel efficiency, vehicle dimensions, etc
- Important: also has CEX auto supplement, links demographics $\leftrightarrow$ new vehicle purchases

Application of welfare from new products: the minivan (Petrin, 2002)

- Main difference to BLP: includes micro-moments using his data from CEX
 - $E[i \text{ buys new vehicle} \mid \text{low income}]$
 - $E[i \text{ buys new vehicle} \mid \text{mid income}]$
 - $E[i \text{ buys new vehicle} \mid \text{high income}]$
 - $E[\text{family size of } i \mid i \text{ purchase minivan}]$
 - $E[\text{family size of } i \mid i \text{ purchase station wagon}]$
 - etc...
- Include these as additional moments in the GMM part of the BLP estimator
- Benefit: more info on heterogeneity $\rightarrow$ can better capture substitution patterns
- Advice: add micro-moments where possible
- (Note: also uses product characteristics as instruments)

Application of welfare from new products: the minivan (Petrin, 2002)

TABLE 4
PARAMETER ESTIMATES FOR THE DEMAND-SIDE EQUATION

Variable	OLS Logit (1)	Instrumental Variable Logit (2)	Random Coefficients (3)	Random Coefficients and Microdata (4)
A. Price Coefficients (α 's)				
α_1	.07 (.01)**	.13 (.01)**	4.92 (9.78)	7.52 (1.24)**
α_2			11.89 (21.41)	31.13 (4.07)**
α_3			37.92 (18.64)**	34.49 (2.56)**

Application of welfare from new products: the minivan (Petrin, 2002)

	B. Base Coefficients (β 's)			
Constant	-10.03 (.32)**	-10.04 (.34)**	-12.74 (5.65)**	-15.67 (4.39)**
Horsepower/weight	1.48 (.34)**	3.78 (.44)**	3.40 (39.79)	-2.83 (8.16)
Size	3.17 (.26)**	3.25 (.27)**	4.60 (24.64)	4.80 (3.57)*
Air conditioning standard	-.20 (.06)**	.21 (.08)**	-1.97 (2.23)	3.88 (2.21)*
Miles/dollar	.18 (.06)**	.05 (.07)	-.54 (3.40)	-15.79 (.87)**
Front wheel drive	.32 (.05)**	.15 (.06)**	-5.24 (3.09)	-12.32 (2.36)**
Minivan	.09 (.14)	-.10 (.15)	-4.34 (13.16)	-5.65 (.68)**
Station wagon	-1.12 (.06)**	-1.12 (.07)**	-20.52 (36.17)	-1.31 (.36)**
Sport-utility	-.41 (.09)**	-.61 (.10)**	-3.10 (10.76)	-4.38 (.41)**
Full-size van	-1.73 (.16)**	-1.89 (.17)**	-28.54 (235.51)	-5.26 (1.30)**
% change GNP	.03 (.01)**	.03 (.01)**	.08 (.02)**	.24 (.02)**

Application of welfare from new products: the minivan (Petrin, 2002)

- Counterfactual exercise: eliminate minivan
- Compute change in consumer welfare ('compensating variation')
- (Note: prices allowed to readjust too using the supply-side)

Application of welfare from new products: the minivan (Petrin, 2002)

TABLE 8
AVERAGE COMPENSATING VARIATION CONDITIONAL ON MINIVAN PURCHASE, 1984:
1982–84 CPI-ADJUSTED DOLLARS

	OLS Logit	Instrumental Variable Logit	Random Coefficients	Random Coefficients and Microdata
Compensating variation:				
Median	9,573	5,130	1,217	783
Mean	13,652	7,414	3,171	1,247

Application of welfare from new products: the minivan (Petrin, 2002)

- CV in logit model is biased upwards (due to large logit draws - i.e. consumers in the model with extreme tastes for minivans)
- Petrin claims: adding the micro-data reduces reliance of the model on the logit draws, and so reduces the overall welfare gain.
- Additionally: directly observes the counterfactual new product choice (so market shares with the new product are correct - we mentioned earlier this was useful in reducing the red-bus blue-bus problem).

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Apple-cinnamon cheerios war

- Hausman paper in “The Economics of New Goods” (1997)
- Values welfare contribution of Apple Cinnamon Cheerios at 60 million dollars per year (in mid 1990s)
- Bresnahan (1997) disagrees with this computation. Criticises identifying assumptions, assumptions about competition, etc...
- Not time to go into it in detail, but on reading list and highly recommended
- Quote from Bresnahan (1997): *‘I have never met an economist who strayed that far from reality by ideology – only by arrogance.’*



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Main takeaways of demand estimation part of this course

- We learned the Mixed Logit ('BLP') model, a model of **demand for differentiated products**.
- **Main components**: product characteristics, observed demographics, unobserved tastes for characteristics, unobserved demand shocks
- Captures **more consumer heterogeneity** than standard logit. An example where this really matters is getting substitution patterns right.
- We saw how to estimate it using the BLP method, common data that are useful or required, and common computational problems
- We studied common instruments (and discussed identification - e.g. recall **dual role** of instruments)
- We saw some applications: measuring conduct, valuing new goods (there are *many* more - often used whenever you need demand in a model)

Main takeaways of demand estimation part of this course

- Computational note: I will get you to code up a simple version of BLP in the homework.
- However, if you actually use it, I highly recommend the PyBLP package (Conlon and Gortmaker). This will save potentially years of computation/coding time!